COURSE: Engineering Design 2

MODULE: GA 3 - Task 1 – PCB CAD Design

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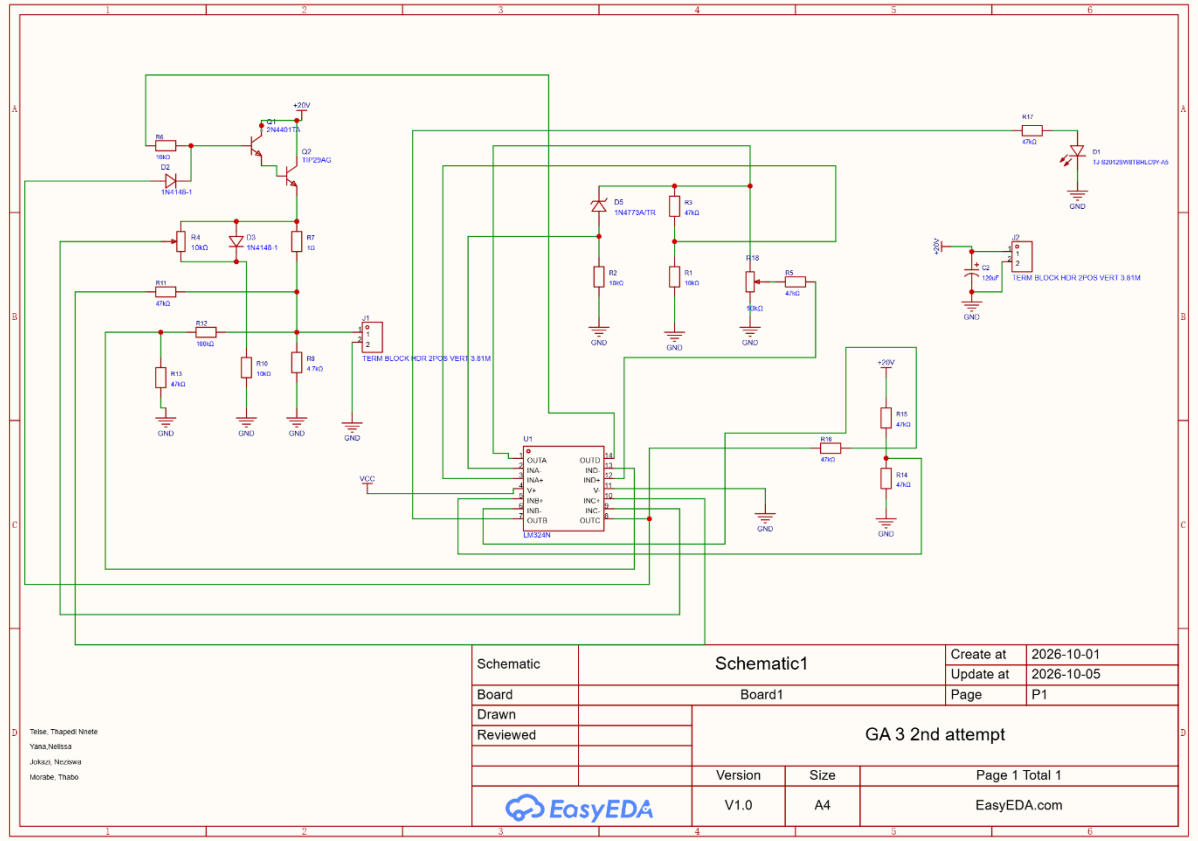
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STUDENT NAME: Thabo Morabe **STUDENT NUMBER:**

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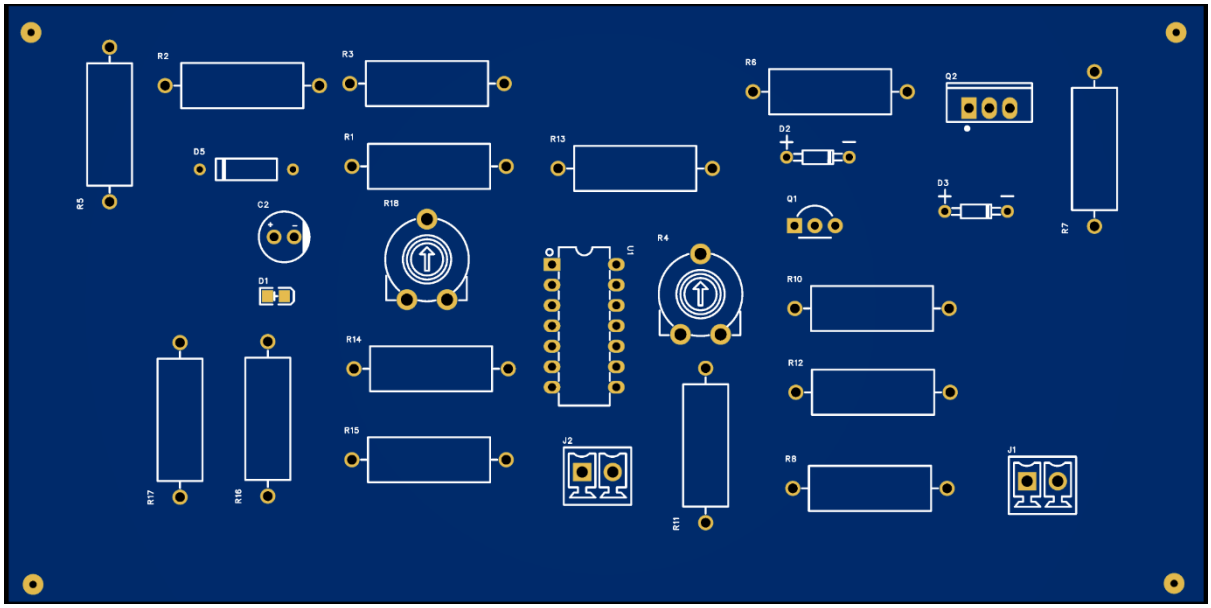
DATE: 05 October 2026

1.Schematic Capture

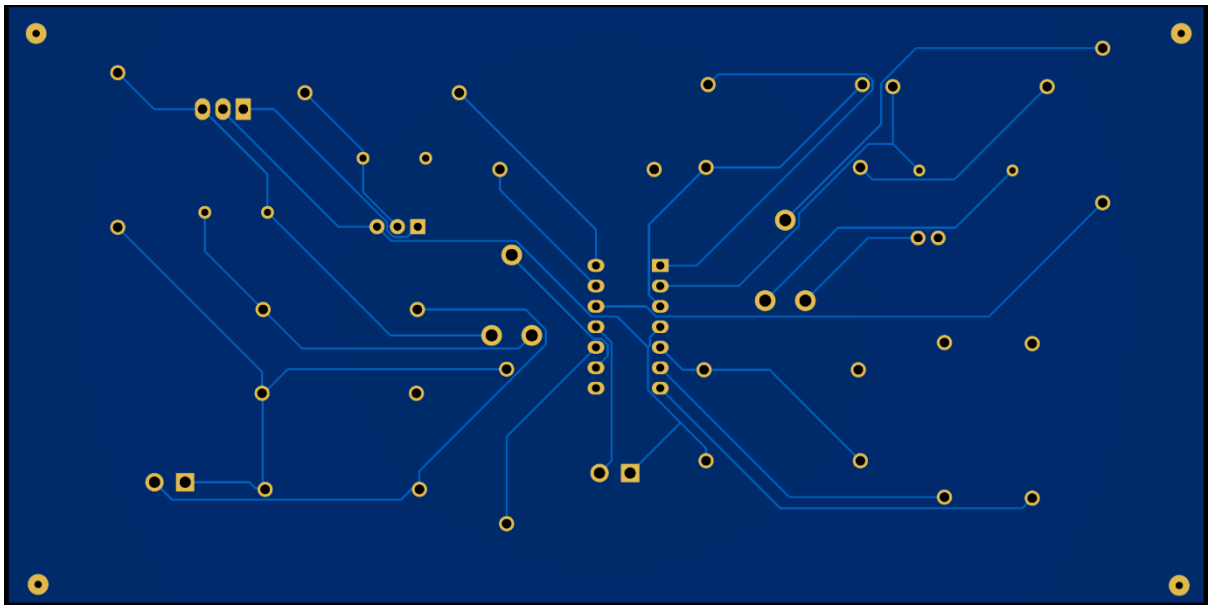


2.PCB Layout

2D Top View

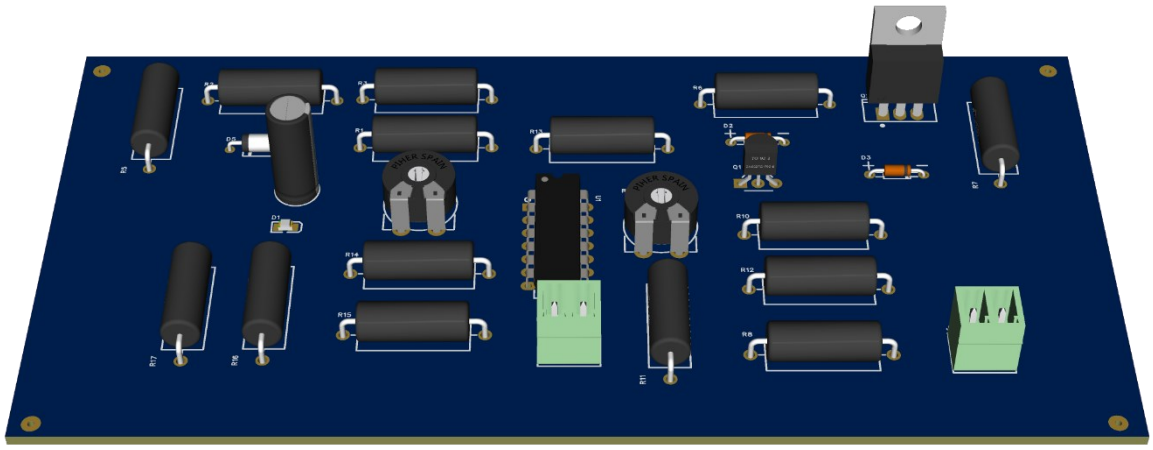


2D Bottom View

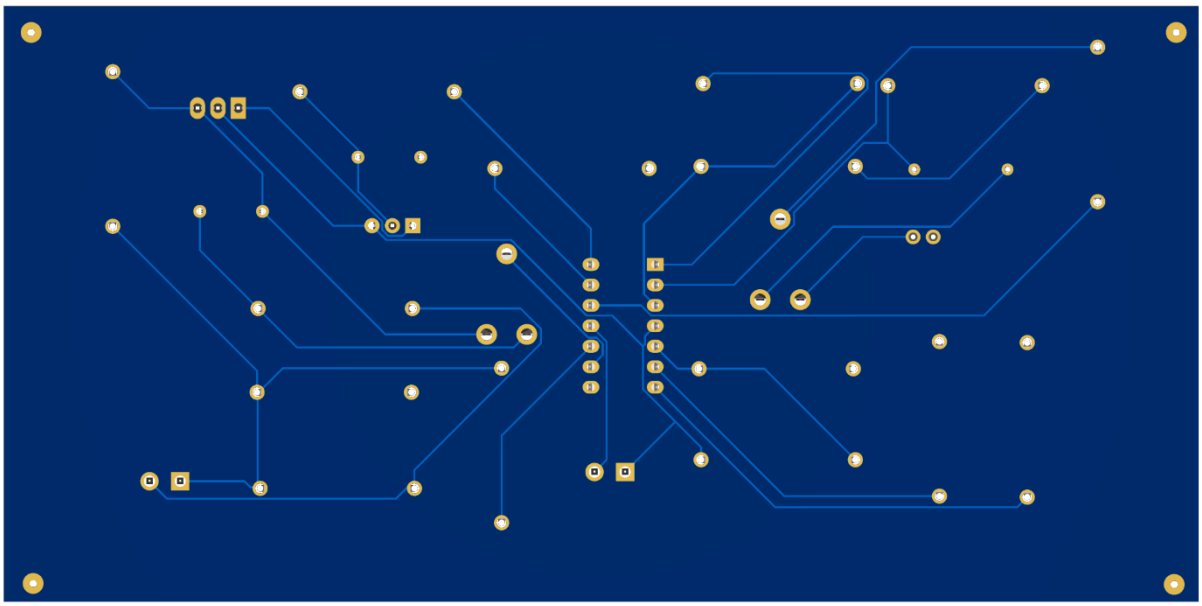


3.3D PCB Views

3D Top View



3D Bottom View



4.Design Notes/Reflection

The problems that were encountered during the completion of the project included firstly understanding what the outcome of the project was supposed to be. There was some confusion at first on whether the group had to assemble an improved circuit which included improving the configuration and replacing some components. Soon it was realised that we had to redistribute the PCB layout in a different way as what was given. Another challenge was building the circuit schematic. The circuit schematic was difficult to build in some sections but eventually after taking time to understand the circuit, the layout could be completed correctly. Routing was also another difficulty as it was the group's first time using EasyEDA, this required a lot of learning about the functions and tools that the software provides, which the group managed to do. In terms of PCB layout, there were challenges with finding the most sensible way of placing components to accommodate current flow and minimise interference.

For improvements, upgrading to a double-sided board would reduce routing complexity and allow for improved signal flow. Adding, clear polarity markings on components so that the viewer can understand the circuit better and it would allow for once the PCB is built that the process is simpler.

In conclusion, the project was a great learning experience. The group learned to design and build a circuit schematic and PCB layout (2D and 3D). Although there were many challenges, the final project objective was achieved. The group is therefore more capable of handling similar projects in the future with more dedication and competency.